Handwritten Digit Recognition Using Convolutional Neural Networks (CNN)

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GitHub Repository:
https://github.com/badar019/cnn-handwritten-digit-recognition

May 17, 2026

1 Introduction

This report presents the design of a Convolutional Neural Network (CNN) for handwritten digit recognition using MATLAB. The system classifies digits (0–9) from grayscale images while improving generalization using augmentation and dropout.

2 Methodology

The workflow includes dataset preparation, model design, and training:

- **Dataset:** Built-in DigitDataset containing 28×28 grayscale images.
- **Split:** 80% training and 20% testing data.
- **Augmentation:** Random rotations ($\pm 10^{\circ}$) and translations (± 3 pixels).
- **CNN Model:** Three convolutional layers, followed by 20% dropout and a fully connected softmax layer.
- **Training:** Adam optimizer with learning rate 0.001 for 8 epochs.

3 Results

3.1 Dataset Samples

Figure 1 shows sample handwritten digits.

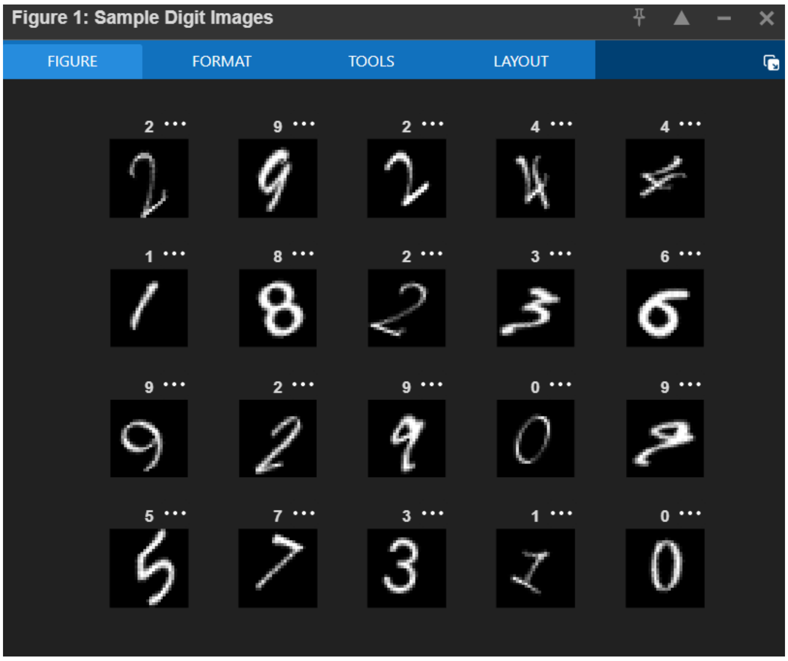

Figure 1: Sample digit images

The dataset shows high variation in handwriting styles, justifying the use of CNN-based feature extraction.

3.2 Model Performance

Figure 2 shows the confusion matrix.

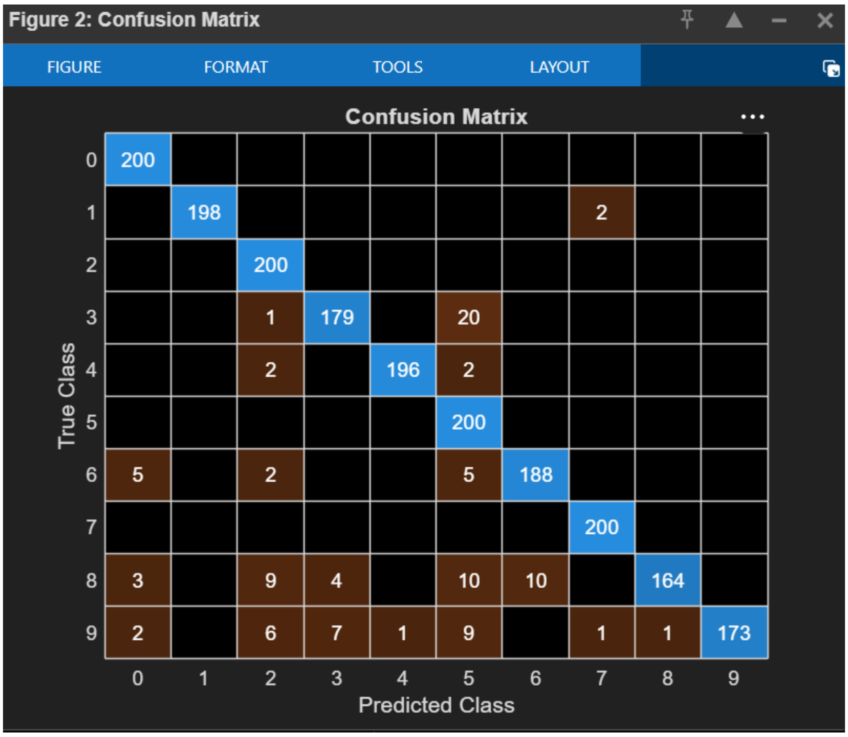

Figure 2: Confusion matrix

The model achieved strong classification performance with high accuracy. Minor confusion occurs between visually similar digits such as 3–5 and 8–9.

3.3 Predictions

Figure 3 shows sample predictions.

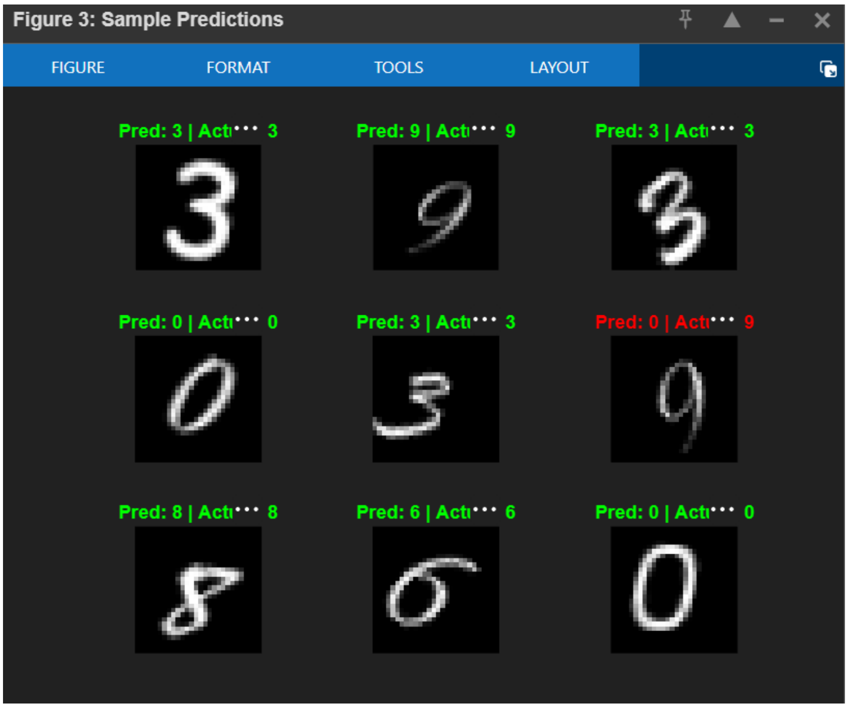

Figure 3: Sample predictions

Most digits are correctly classified, with occasional misclassification due to unclear stroke patterns.

4 Conclusion

The CNN model effectively recognizes handwritten digits with high accuracy. Data augmentation and dropout improve robustness and reduce overfitting. Future improvements may include deeper architectures and learning rate scheduling.